WEEK 7 - HEMATOLOGY • SAL HISTOPATHOLOGY LAB • PRE-SAL DRAFT

Blood & Bone Marrow Histopathology

Summary-based notes now • update with the actual SAL slides after class

Official lab LOs | Summary-based | Visual recognition first | 3 ScholarRx Bricks | Not core lecture numbered

PRE-SAL VERSION. This is intentionally built before the in-class SAL slide set is released. It uses the official Week 7 lab objectives, the existing Peripheral Blood & Bone Marrow Histology summary, and cross-references from this week's Hematopoiesis / Anemia / Hemolysis material. After SAL, replace the borrowed reference images and add the instructor's exact specimens, labels, and emphasis.

Official Lab Objectives

1. Recognize normal peripheral-blood RBC histology and identify anisocytosis, microcytosis, spherocytosis, sickled cells, and reticulocytosis.

2. Recognize normal histology and functions of lymphocytes, neutrophils, eosinophils, basophils, and monocytes.

3. Recognize normal peripheral-blood platelets.

4. Recognize normal bone marrow and differentiate a bone-marrow smear from a peripheral-blood smear.

SCHOLARRX ASSIGNED

Histology of Blood Vessels & Lymphatics
Structure & Function of Bone Marrow
Erythrocyte Laboratory Tests

THE SAL QUESTION TO ASK EVERY TIME

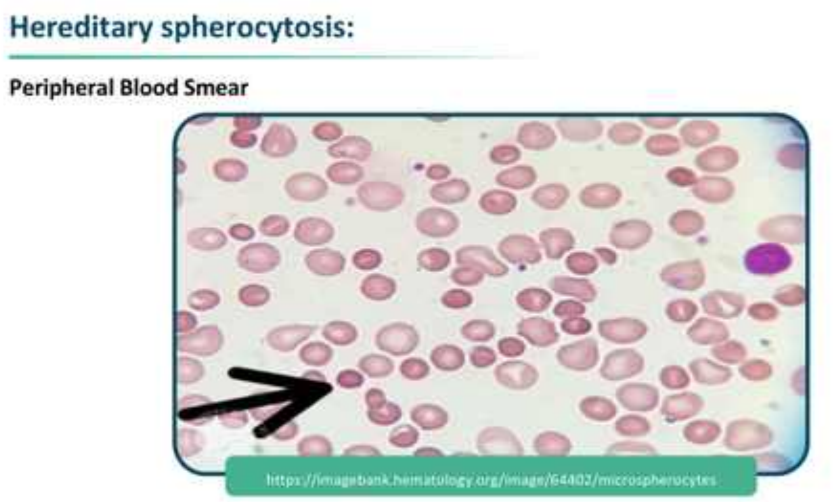
What am I looking at: mature peripheral blood, or a marrow full of developing precursors? Then identify the lineage/cell by **size, nucleus, cytoplasm, granules, and central pallor.**

RBC Recognition

Finding	What it looks like	What it means / attach mentally
Normal RBC	Uniform round biconcave disc; anucleate; pink with central pallor about one-third of diameter.	Use normal neighbors as the size/ color reference for everything else.
Anisocytosis	Noticeable cell-to-cell variation in RBC size.	Think increased RDW / mixed RBC population. Morphology word = *size variability*.
Microcytosis	RBCs smaller than expected, often with increased central pallor when hypochromic.	Iron deficiency / thalassemia framework. Do not diagnose etiology from size alone.
Spherocytosis	Round, dense RBC with **little or no central pallor**.	Membrane surface-area loss. Classic HS; can also appear in warm AIHA.
Sickled cell	Elongated crescent / boat-shaped RBC with pointed ends.	HbS polymerization under deoxygenated conditions.
Reticulocytosis	Polychromasia: slightly larger, bluish-gray RBCs on Wright-Giemsa because residual RNA remains.	Marrow response to anemia / blood loss / hemolysis.

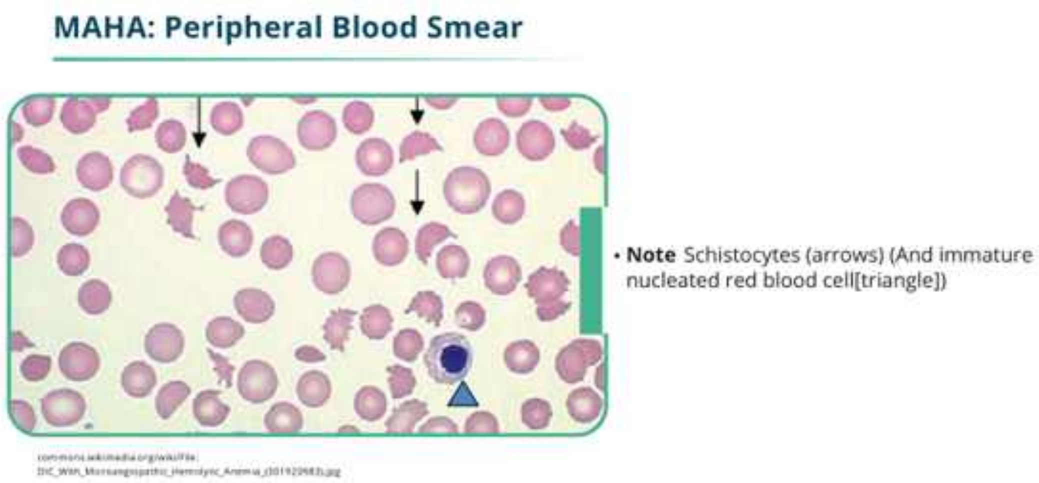
SAL TRAP: SPHEROCYTE VS MICROCYTE

Both can look “small.” A **spherocyte is dense and lacks central pallor**; a typical microcyte is small but still shows central pallor, often exaggerated if hypochromic.



Cross-reference from Week 7 Hemolytic Anemias, not an SAL slide. Use this only as a pre-SAL morphology anchor for spherocytes. The actual lab image should replace it after class.

Useful Week 7 extras, not explicit lab-LO requirements: schistocytes = fragmented/helmet RBCs in MAHA; target cells = excess membrane-to-volume appearance; bite cells = oxidant injury/G6PD. These are worth recognizing because the week's disease lectures repeatedly use smear morphology.



Cross-reference from Week 7 Hemolytic Anemias, not an SAL slide. This is an extra cross-reference because schistocytes are repeatedly tested elsewhere in Week 7 even though they are not explicitly named in this SAL objective.

SAL IMAGE SLOT: normal RBCs • anisocytosis • microcytosis • sickled cells • reticulocytosis
Replace/add the class specimens here after SAL.

Peripheral WBC Recognition

Cell	Morphology	Main function / clinical association
Neutrophil	2-5 nuclear lobes connected by thin filaments; pale cytoplasm with fine granules.	Rapid phagocyte for acute bacterial inflammation. Band = U/horseshoe nucleus, immature neutrophil.
Eosinophil	**Bilobed nucleus + large red/orange granules.**	Helminth defense, allergy/asthma biology; granules contain major basic protein.
Basophil	Large coarse dark blue-purple granules often **obscure the nucleus**.	IgE-linked allergic/inflammatory mediator release; histamine/heparin association.
Lymphocyte	Usually small; round very dark nucleus occupies most of cell; thin rim of pale blue cytoplasm.	B, T, and NK lineages cannot be reliably distinguished by routine morphology alone.
Monocyte	**Largest normal circulating WBC**; folded/kidney-bean nucleus; gray-blue “ground-glass” cytoplasm, sometimes vacuolated.	Phagocyte/APC precursor that enters tissues and differentiates into macrophages.

FIVE-CELL SPEED ID

Neutrophil = segmented. Eosinophil = orange + bilobed. Basophil = purple granules hide nucleus. Lymphocyte = nucleus almost fills cell. Monocyte = biggest + kidney/folded nucleus.

Ozzy checkpoint - click him once you can identify all five WBCs without reading the table.

BAND VS MONOCYTE

A band neutrophil has a relatively uniform **U-shaped/horseshoe nucleus** with neutrophilic granules. A monocyte is larger, with a folded/kidney nucleus and more abundant gray-blue often vacuolated cytoplasm.

SAL IMAGE SLOT: neutrophil • band • eosinophil • basophil • lymphocyte • monocyte

Platelets

What to recognize on peripheral smear

- Very small, anucleate **purple-blue cytoplasmic fragments**, much smaller than an RBC.
- Can appear singly or in small aggregates.
- Derived from marrow megakaryocytes; not complete cells.
- Central darker granulomere contains alpha and dense granules; peripheral hyalomere contains cytoskeletal elements.

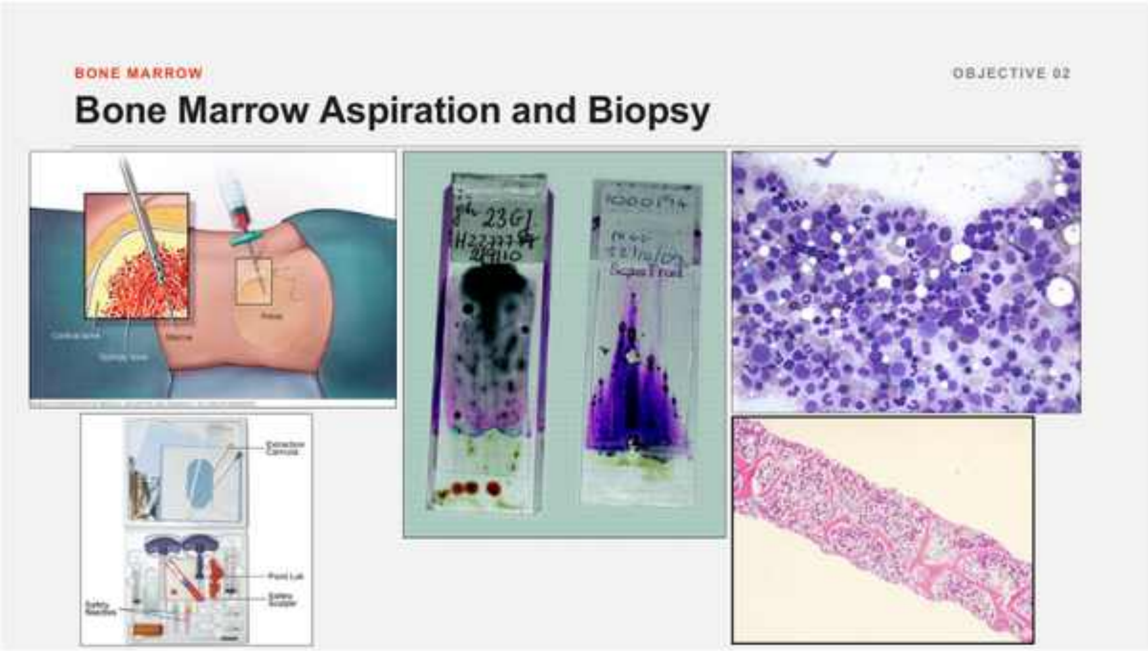
Granule	Attach mentally
Alpha	vWF, fibrinogen, PDGF and other proteins
Dense / delta	ADP, Ca^{2+}, serotonin

FUNCTION CROSS-LINK

Adhesion: vWF-GpIb -> activation/degranulation -> aggregation: GpIIb/IIIa-fibrinogen.

SAL IMAGE SLOT: normal platelet morphology and size comparison to RBCs

Bone Marrow Histology & Smear Recognition



Cross-reference from Week 7 Hematopoiesis & Lymphoid System, not an SAL slide. Cross-reference for the difference between aspirate and core biopsy. The SAL may use either smear morphology or intact marrow architecture.

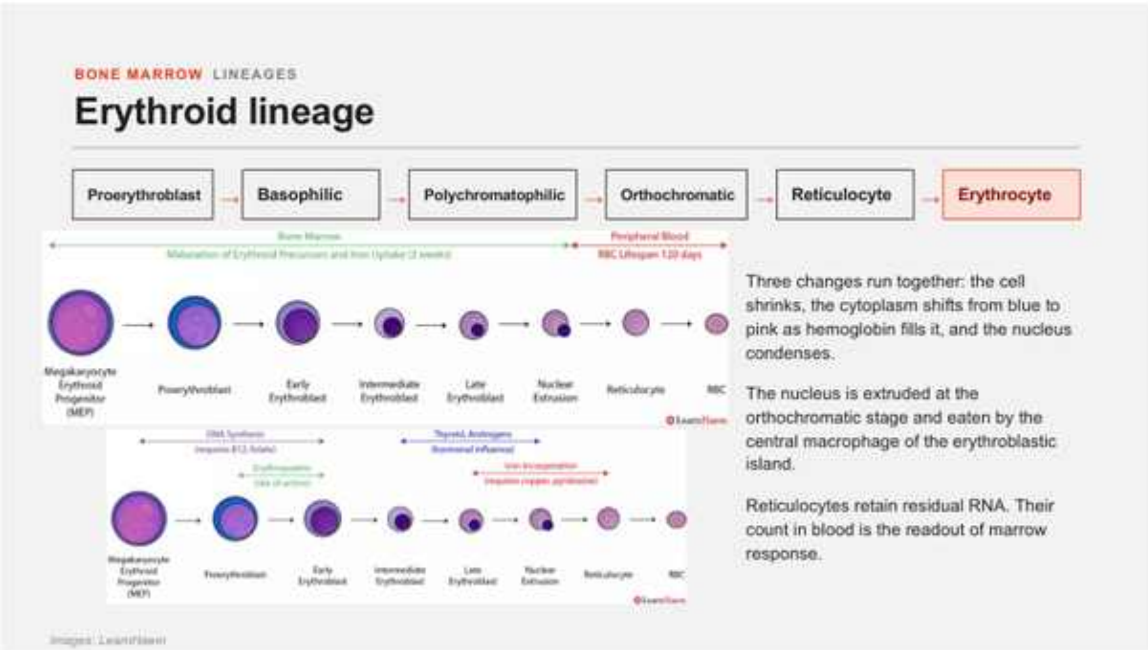
	Peripheral blood smear	Bone marrow aspirate/smear
Background	Mostly mature RBCs with scattered mature WBCs + platelets	Very cellular; many nucleated precursors at different maturation stages
Erythroid cells	Normally mature anucleate RBCs; retics may be seen as polychromasia	Nucleated erythroblasts from early to late maturation
Myeloid cells	Mostly mature segmented cells	Myeloblast -> promyelocyte -> myelocyte -> metamyelocyte -> band -> mature forms
Megakaryocytes	Absent; only their platelet fragments circulate	**Huge multilobulated cells**, often obvious
Other clues	No normal marrow spicules/fat architecture	May see marrow spicules, adipocytes, mixed precursor populations

Cross-reference from Week 7 Hematopoiesis & Lymphoid System, not an SAL slide. Cross-reference for the normal mixed marrow environment: megakaryocytes, erythroid islands, myeloid precursors, adipocytes, and bony trabeculae.

IMMEDIATE MARROW GIVEAWAY

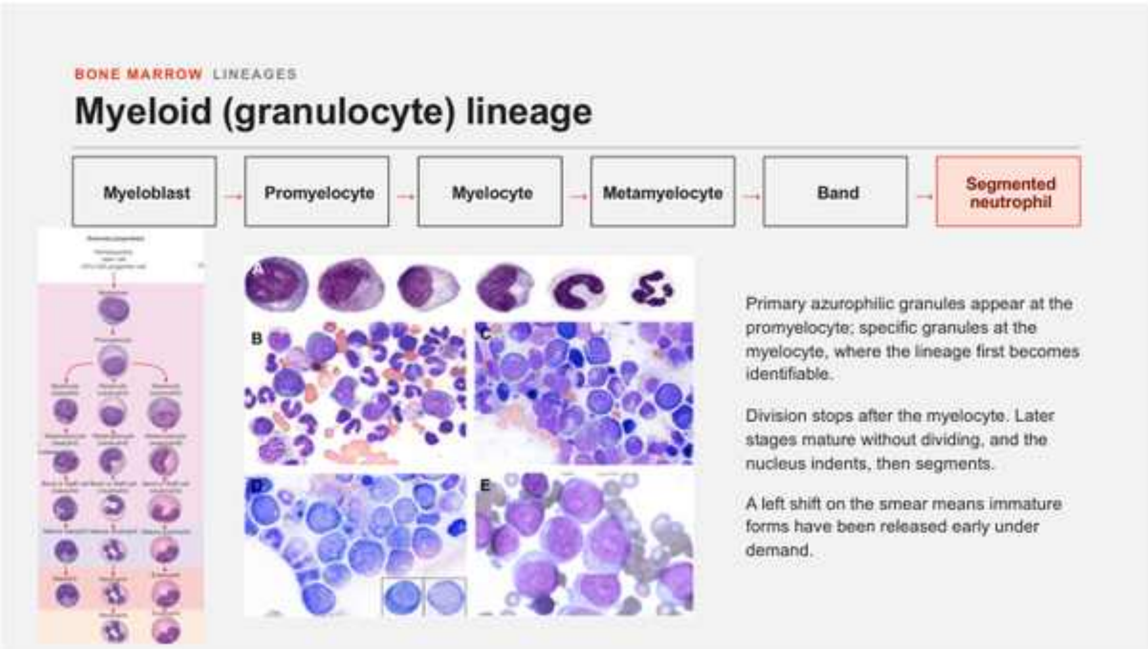
A giant multilobulated megakaryocyte + many nucleated precursor stages = marrow, not peripheral blood.

Erythroid maturation



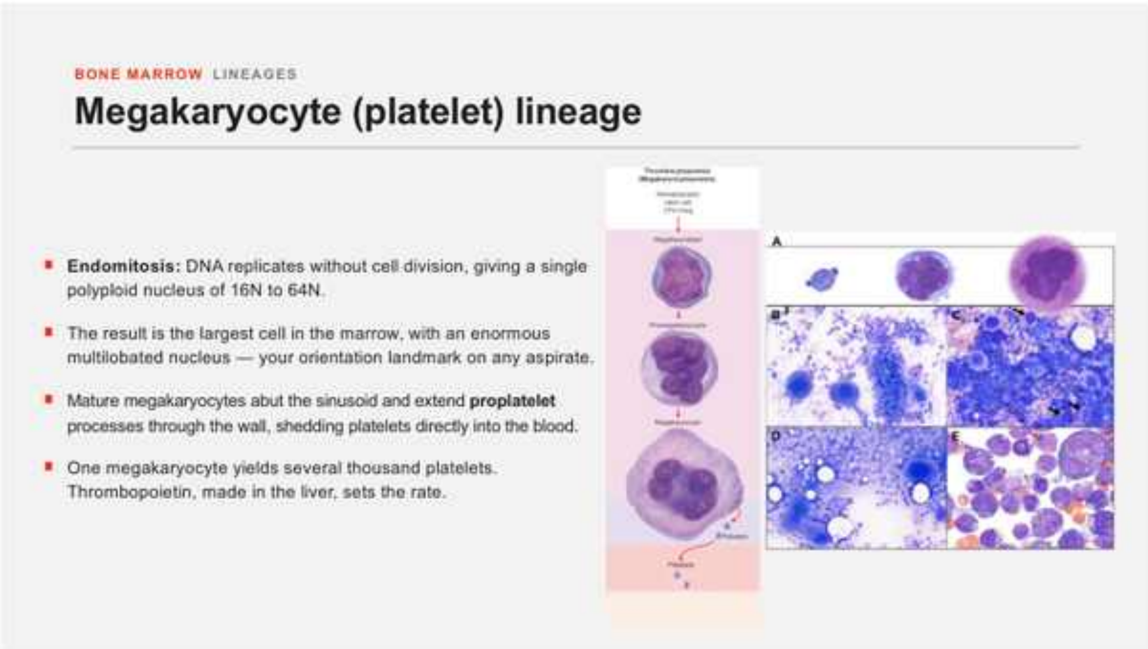
Cross-reference from Week 7 Hematopoiesis & Lymphoid System, not an SAL slide. Cell gets smaller; cytoplasm goes blue -> pink as hemoglobin accumulates; nucleus condenses and is ultimately extruded.

Granulocyte maturation



Cross-reference from Week 7 Hematopoiesis & Lymphoid System, not an SAL slide. Myeloblast -> promyelocyte -> myelocyte -> metamyelocyte -> band -> segmented neutrophil.

Megakaryocyte lineage



Cross-reference from Week 7 Hematopoiesis & Lymphoid System, not an SAL slide. Megakaryocytes become huge and polyploid by endomitosis, then shed platelets toward marrow sinusoids.

SAL IMAGE SLOT: actual in-class marrow smear(s), normal marrow core/ biopsy, and any instructor-labeled precursors

Fast SAL Identification Algorithm

#	Ask	If yes...
1	**Is this field mostly uniform anucleate pink discs?**	Peripheral blood. Start with RBC morphology, then find WBCs/platelets.
2	**Are there many nucleated cells at multiple developmental stages?**	Think bone marrow.
3	**Is there a giant multilobulated cell?**	Megakaryocyte -> marrow.
4	**For an RBC abnormality, is the issue size, pallor, shape, or color?**	Size = anisocytosis/microcytosis; no pallor = spherocyte; crescent = sickle; bluish larger RBC = retic/polychromasia.
5	**For a WBC, what is the nucleus doing?**	Segmented neutrophil; bilobed eos; obscured basophil; round dense lymphocyte; kidney/folded monocyte.

WHAT CHANGES AFTER SAL

The **content framework stays**. What we will replace/add is the exact set of instructor images, the labels they point out, which cells they expect you to distinguish at low vs high power, and any specific lab practical traps.

Objective Review

LAST 60 SECONDS

Normal RBC: uniform with central pallor. **Anisocytosis:** variable sizes. **Microcyte:** small. **Spherocyte:** dense/no pallor. **Sickle:** crescent. **Retic:** larger/bluish.
WBCs: segmented neutro, orange bilobed eos, purple-granule baso, round-dark lymphocyte, big kidney-nucleus monocyte.
Platelets: tiny purple fragments.
Marrow: many nucleated precursors + giant megakaryocytes + mixed maturation; blood = mostly mature cells.